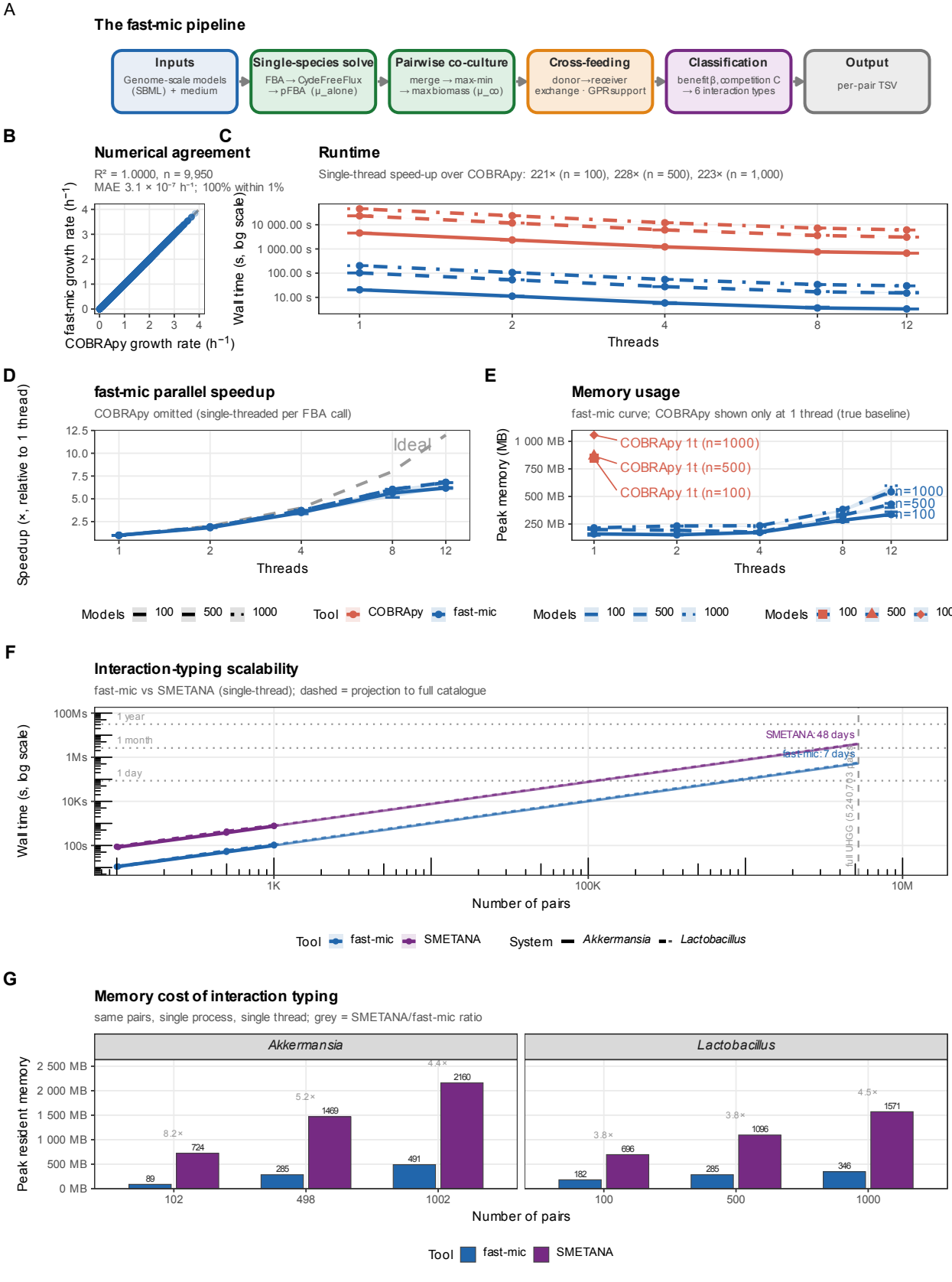

A
The fast-mic pipeline
Inputs
Genome-scale models (SBML) + medium
Single-species solve
FBA → CycleFreeFlux → pFBA (μ_alone)
Pairwise co-culture
merge → max-min → max biomass (μ_co)
Cross-feeding
donor→receiver exchange · GPR support
Classification
benefit β, competition C → 6 interaction types
Output
per-pair TSV
B
Numerical agreement
R² = 1.0000, n = 9,950
MAE 3.1 × 10⁻⁷ h⁻¹; 100% within 1%
fast-mic growth rate (h⁻¹)
COBRApy growth rate (h⁻¹)
C
Runtime
Single-thread speed-up over COBRApy: 221× (n = 100), 228× (n = 500), 223× (n = 1,000)
Wall time (s, log scale)
10 000.00 s
1 000.00 s
100.00 s
10.00 s
Threads
D
fast-mic parallel speedup
COBRApy omitted (single-threaded per FBA call)
Speedup (×, relative to 1 thread)
Ideal
Threads
Models 100 500 1000
Tool COBRApy fast-mic
E
Memory usage
fast-mic curve; COBRApy shown only at 1 thread (true baseline)
Peak memory (MB)
1 000 MB
750 MB
500 MB
250 MB
COBRApy 1t (n=1000)
COBRApy 1t (n=500)
COBRApy 1t (n=100)
n=1000
n=500
n=100
Threads
Models 100 500 1000
Models 100 500 1000
F
Interaction-typing scalability
fast-mic vs SMETANA (single-thread); dashed = projection to full catalogue
Wall time (s, log scale)
100Ms
1Ms
10Ks
100s
1 year
1 month
1 day
SMETANA: 48 days
fast-mic: 7 days
full UHGG (5,240,703 pairs)
1K
100K
10M
Number of pairs
Tool fast-mic SMETANA
System Akkermansia Lactobacillus
G
Memory cost of interaction typing
same pairs, single process, single thread; grey = SMETANA/fast-mic ratio
Peak resident memory
Akkermansia
Lactobacillus
2 500 MB
2 000 MB
1 500 MB
1 000 MB
500 MB
0 MB
89
724
8.2×
285
1469
5.2×
491
2160
4.4×
182
696
3.8×
285
1096
3.8×
346
1571
4.5×
102
498
1002
100
500
1000
Number of pairs
Tool fast-mic SMETANA